Motorbike Helmet

Paraffin Wax PCM

Group 24

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Subject Name:

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Overview of Work Through Four Sprints

The following section of the report will outline and summarise all the activities completed for each of the four Sprints in this subject.

Sprint 1

Our primary focus on Sprint 1 was on identifying and defining the engineering problem related to heat retention in motorbike helmets. Below is a summary of the key points covered in this Sprint:

1. **Defining the Engineering Problem:** The challenge was to acknowledge how motorbike helmets effectively dissipate heat energy, leading to discomfort for riders. Key design requirements were considered such as safety, weight, comfort, and preventative/corrective maintenance.
2. **Proposed Design Solution:** We were able to identify a range of different PCMs, both organic and inorganic, that may have provided a potential solution to address the heat retention issue in motorbike helmets. PCM offers efficient absorption and dissipation of thermal energy by transforming from a solid to a liquid.
3. **Investigation & Evaluation of PCM Impact:** The potential impacts of integrating PCM into motorbike helmets, focusing on factors such as weight, comfort, and safety were analysed. Thorough research was conducted, as shown in the section Specific Case Study Presentation. The findings highlighted the benefits of organic paraffin PCMs in delaying temperature increases and improving general comfort for motorbike riders.
4. **Review of Existing Studies:** Relevant research articles were reviewed to gain insights into PCM cooling systems for motorbike helmets. These studies provided invaluable information that assisted in determining the PCM selection criteria, integration methods, and the effectiveness of PCM motorbike helmets.
5. **Identification of Design Challenges:** Various challenges were identified that were associated with integrating PCM into helmets, such as material selection, heat distribution, and structural integrity. Proposed solutions to these issues included thorough testing, model design optimisation, and rapid prototyping.

Sprint 2

For Sprint 2, our key area of focus shifted to the meshing analysis and integration of the paraffin PCM into our motorbike helmet model. Below is a summary of the tasks that were covered in this Sprint:

1. **CAD Model:** A CAD model for our PCM helmet was obtained for modification. We modelled the helmet geometry to incorporate the PCM layers, followed by a meshing analysis, in ANSYS, as part of the simulation process.

2. **Section of Helmet Investigated:** The section of the helmet was clearly outlined for further analysis and meshing. This allowed us to focus our approach on a particular part of the motorbike helmet CAD model.
3. **Individual Domain Layers:** Different layers within the helmet design were examined, and we considered materials such as Expanded Polystyrene (EPS) and Polyurethane (PUR) for thermal absorption and distribution.
4. **Ambient Heat/Flow Conditions:** Numerous heat transfer scenarios were analysed, including adiabatic insulation conditions and ambient heat/flow conditions, to understand how different environmental factors affect the cooling effectiveness of the helmet.
5. **Meshing Analysis:** Meshing was performed at different sizes, ranging from coarse to very fine meshes, to investigate how computational efficiency changes with each size. We were also able to determine which sizes would provide us with the most accuracy in our simulations while maintaining efficiency in processing.

Sprint 3

Sprint 3 mainly focussed on the simulation and analysis for the meshing quality and convergence. Here's a breakdown of the key points covered in this presentation:

1. **Material Properties & PCM Layers:** We defined the material properties of the PCM (Paraffin) and its surrounding layers, including specifications and specific heats, which were sourced from reliable references.
2. **Boundary Conditions:** Boundary conditions were precisely defined based on our research findings, to ensure realistic simulation environments for accurate results.
3. **Mesh Quality & Convergence Analysis:** Various mesh qualities were utilised for our simulations. The convergence analysis was performed, allowing us to assess the accuracy and stability of the results.
4. **Simulation Results:** The results from our simulations were presented. Some details that were presented included time for PCM melting, volume average liquid fraction graphs, and convergence residual graphs for each mesh size (coarse, fine, and very fine).

Sprint 4

Sprint 4 covered the rapid prototyping and post-processing results for our motorbike PCM helmet. Below is a breakdown of the key points covered in this Sprint:

1. **Rapid Prototyping – 3D Printing:** The final prototypes for the helmet geometry and the domain layers were fabricated via 3D printing, utilising the insights gained from simulation and analysis in Sprint 3. The final version of the 3D-printed helmet was presented, along with the specifications and settings used for 3D-printing.
2. **Post-Processing Results:** Results from the simulation and post-processing were detailed, including temperature distribution and liquid fraction. Boundary conditions were considered, and contour analysis offered an understanding of temperature distribution and liquid fraction contours.

Background Explanation

Motorcycle helmets are an indispensable tool for safety but come with the consequence of often causing thermal discomfort to the wearer's head, particularly in hot climates or during periods of extended use. This discomfort results from the accumulation of metabolic heat and limited ventilation which leads to elevated head temperatures.

Our research aims to build upon prior research as well as analysis of several other academic articles with the final goal of our project aims to advance the development of PCM-based helmet cooling systems. We will investigate the integration of new PCMs directly into the helmet liner and utilise 3D-printed PCM structures as well as Ansys simulations for testing and simulating optimal heat transfer. The overarching goal of our research is to develop a practical, efficient, and sustainable PCM-based helmet cooling solution that significantly enhances rider comfort and safety.

Phase Change Materials (PCMs) are a class of materials that undergo a phase transition such as a solid-to-liquid state or a liquid-to-solid state when they reach a specific temperature range. During these transitions, the PCM absorbs or releases a substantial amount of latent heat and thermal energy thus making them ideal for use as thermal energy storage devices. For this example, we will focus on helmet cooling in which case the PCM can absorb excess heat from the wearer's head and change liquid state, cooling the wearer's head. Research by Tan and Fok (2006) demonstrated the feasibility of using PCMs for helmet cooling, reporting significant reductions in head temperature and delayed onset of thermal discomfort.

Paraffin waxes are a specific type of solid-liquid PCM are suitable for use in this due to favourable properties. Paraffins are organic compounds derived from petroleum or natural sources they have a high latent heat of fusion meaning they can store a large amount of thermal energy per unit mass during melting. This translates to being able to cool efficiently with a lower material volume which is an essential consideration for wearable devices like helmets. Moreover, paraffin waxes are available with a wide range of melting points which allows for customization of the cooling effect based on the desired temperature range for optimal comfort. Furthermore, their chemical stability and relative non-toxicity further ensure safe use near the human body. Finally, Paraffin waxes are relatively inexpensive and readily available, making them a practical choice for large-scale production.

To integrate paraffin wax PCMs into helmets, various techniques have been explored. In this regard, we'll consider microencapsulation which involves enclosing the PCM in tiny, protective shells that prevent leakage and facilitate even distribution within the helmet liner. Additionally, we'll explore how advancements in 3D printing technology have opened possibilities for creating complex structures with embedded PCMs, enabling customised designs that maximise heat transfer efficiency and wearer comfort.

The first paper investigated in this research is *A Thermal Test System for Helmet Cooling Studies* (Fitzgerald et al, 2018). The paper assesses the cooling effectiveness of bicycle helmets by using a heated plastic thermal head form. The construction consists of a 3D-printed head form of low thermal conductivity with an internal layer of high thermal mass that is heated to a constant uniform temperature by an electrical heating element. Testing of the thermal head was conducted on the bare exposed head (BH) as well as four bicycle helmets, a Time Trial helmet (TT), an older-generation road cycling helmet (VR), and two modern aero-road helmets designed for both speed and thermal cooling (AR1 & AR2). Thermal testing was run for 50 minutes with an ambient temperature of 38 °C. The value of the paper comes from allowing us to understand what parts of the head are most likely to experience the highest range of temperature thus allowing us to determine where best to place the PCM material.

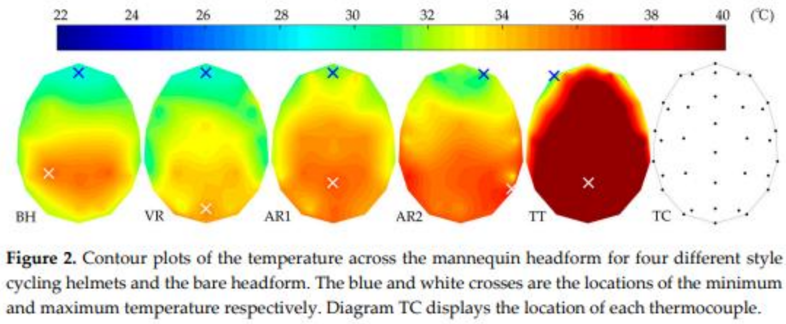


Figure 1. Thermal imaging of the head.

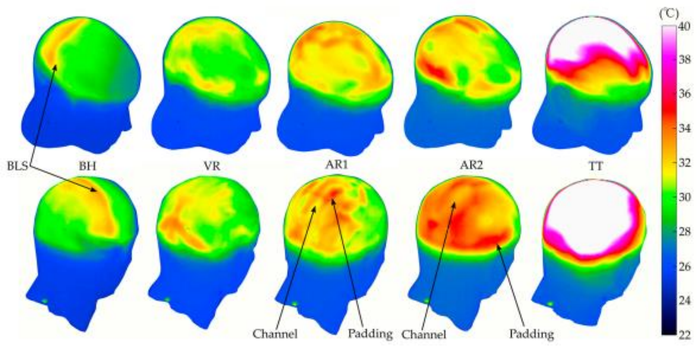


Figure 2. Thermal imaging of head - 3D.

The second article to be investigated is *Helmets cooling with phase change materials: A systematic review* (Adil A.M. et al., 2023). The article is a systematic theoretical and comparative study of research articles and various methods that were used to cool helmets using phase change studies and the value of the paper is in offering a concise comparative study of various other research articles.

(Adil A.M., et al., 2023) Examines the physical properties of the PCM stating the ideal PCM for use in a helmet should have appropriate phase transition temperature, large thermal conductivity, high latent heat capacity, high specific heat, low level of super-cooling, low level of phase separation, and low vapour pressure. The chemical criteria the PCM must have include compatibility with the container material, chemical stability, complete reversible phase change cycle, low flammability, and non-toxicity. A thorough assessment of previous research and summarisation of the conclusions reached by each research paper. E.g.:

Table 3
Summary of the previous studies on helmets with PCMs.

Authors	Study type	Type of helmet	PCM used	PCM encapsulation/location	Main results
Ali et al. [54]	Experimental study	Safety helmet	Paraffin wax with graphene nano-particles	Macro-multilayer nylon pouch/ between the helmet and the head	• The time required to reach comfort zone of 28–33 °C was reduced as the concentration of nano-particles increased. • The PCM could give thermal comfort for >4 h for all concentrations at an ambient temperature of 35 °C.
Al-Rjoub et al. [42]	Experimental study	Brain cooling system	Ice	Macro/heat exchanger	• The cooling effect prolonged as the thermal load decreased. • The helical coil had higher heat transfer rate than straight pipe.

Figure 3. Summary of literature study. (Adil A.M., et al., 2023).

Investigates methods of integrating PCM into helmets including:

- Passive cooling - using thermal changes based on ambient and skin temperature to melt the PCM.
- Active cooling - hybrid systems using PCM alongside active cooling methods such as the use of fans.
- Effect of environmental parameters on the use of PCM
- PCM with Peltier module - module constructed with two types of semiconductor elements between copper substrate making it possible to change the surface temperature.
- Helmets with modified liners to improve comfort and heat dissipation.
- Use of nanoparticles in a typical PCM to increase thermal conductivity.

While the research does not have any practical applications, its value comes from giving us the properties and avenues that need to be investigated when integrating PCM into helmets as well as providing a valuable literature review of existing applications.

The third article to be investigated is *Cooling of helmet with phase change material* (F.L. Tan, S.C. Fok, 2006), this article is one of the earliest discussions of a design for a motorcycle helmet cooling system. The PCM pouch is placed between the helmet and the wearer's head and absorbs heat transferred by conduction through a heat collector. The paper presents a theoretical design of a PCM-cooled helmet designed to maintain the head skin temperature at about 30 °C. It is designed to be capable of cooling the head continuously for up to 2 hours which is referred to as the loading time. After the loading time, the stored heat in the PCM pouch needs to be dissipated to the ambient which is referred to as the discharging time. Since the head skin temperature of the wearer is designed to maintain at around 30 °C, the PCM, Climsel C28, which has a melting temperature of 28 °C is chosen. The helmet can provide cooling for up to two hours, after which the PCM needs to be discharged by immersing it in water for about 15 minutes. Assuming all the heat from the ambient and the head is absorbed by the PCM over a loading time of 2 hours, the amount of PCM required is determined by calculations to be about 903 g and the volume occupied by the PCM is about 618 cm3.

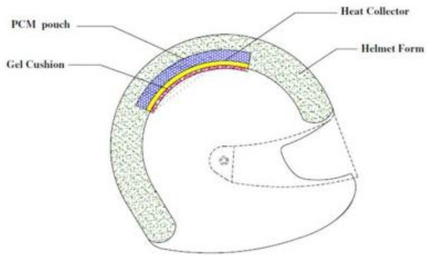


Figure 4. Helmet diagram. (F.L. Tan, S.C. Fok, 2006).

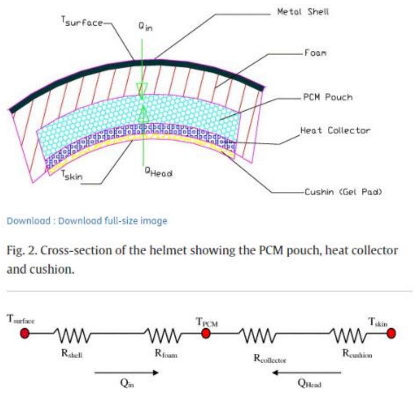


Figure 5. Cross section. (F.L. Tan, S.C. Fok, 2006).

Improving thermal comfort of motorcycle helmet by using phase change materials - (Sharifpour, E. et al., 2012) presents the theoretical design of a helmet cooling system using phase change material (PCM) to absorb and store the heat produced by the rider to achieve comfort cooling for the wearer. It also models the melting process of the PCM, the physical properties of the liquid and solid phases and the heat transfer in both solid and liquid layers.

The physical model of the whole melting process can be divided into three stages: the pre-melt stage through which the heat transfer within the slab is that of pure conduction in a solid material, the melting stage when the solid-liquid interface temperature is equal to the melting temperature of PCM and the post melt stage.

The results show that with the use of PCM, the chosen PCM being Calcium chloride hexahydrate, the temperature inside the helmet can be maintained at a lower temperature for a longer period and illustrates that increasing the thickness of the PCM section improves the period of melting of PCM and decreases the deviation of head skin temperature from the melting temperature of PCM.

The final paper *helmet cooling system using phase change material for long drive.* (A & Hariram, V., et al., 2015). Explores potential practical design for a helmet. Helmet Components:

- **Visor:** Made of strong, transparent material (e.g., polycarbonate) to protect the rider's face from wind, dust, and insects.
- **Outer Shell:** constructed from thermoplastics or fibre-reinforced polymer (FRP) to absorb and disperse impact during accidents.
- **EPS Liner:** Lightweight and highly impact-absorbing expanded polystyrene (EPS) located inside the shell.
- **Comfort Padding:** Firm synthetic foam pad covered with skin-friendly fabric for wearer comfort.

This work mainly focuses on absorbing the heat produced inside the helmet. To achieve this, a suitable Glauber salt PCM has been encapsulated inside the helmet within aluminium foil.

The design featured holes drilled on the front and rear sides of the helmet allowing fresh air to continuously flow in and out of the helmet so that the heat produced in the helmet is instantaneously tapped out. By providing holes at both the front and rear sides of the helmet for forced convective heat transfer through air thus continuous cooling is achieved till the entire PCM fuses.

The paper investigates the use of Computational Fluid Dynamics (CFD) using a numerical method to solve for how long the implemented PCM will cool the wearer using temperature as a boundary condition. Discretization is used to develop approximations of the governing equation and the domain is discretised into a finite set of control volumes or cells known as a mesh.

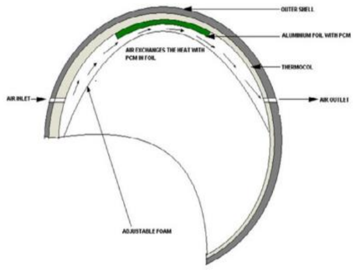


Figure 6. Helmet design. (A & Hariram, V., et al., 2015).

B. Calculation

From the data, Heat flux	$= 22 W/m^2$
Latent heat of PCM	= 251KJ per Kg
From diagram, area	$= 0.05\ m^2$
Latent heat of PCM	= 251000 J/Kg
Therefore for $0.05\ m^2$ area, 0.05 * 22.42	
From diagram, mass	= 0.02Kg
	= 22.42 * 5/100
Therefore for 0.02Kg,	
251000*0.02	= 1.121 watts
Total amount of heat	=5020 joules
Which means, for 1 second, PCM absorbs 1.121 Joules	
From these data	
Time taken for fusion	= 5020/1.121
	=4478 sec
	=1.24hrs

Figure 7. CFD results.

Specific Case Study Presentation

One key case study that examines the use of paraffin PCMs in motorbike helmets is "Integration of Phase Change Material in Helmets" by Shenoy S., published in the International Research Journal of Engineering and Technology (IRJET) in 2018.

This case study explores integrating phase change materials, specifically paraffin-based PCMs, into motorcycle helmets to enhance thermal comfort for motorbike riders. This study investigates the details of heat transfer within motorbike helmets, potential issues in integrating PCM material, and other innovative aspects of the overall design of PCM motorbike helmets.

The study examines the thermal properties of paraffin wax PCMs, including their high latent heat storage capability and appropriate melting temperatures, which make them ideal solutions for thermal regulation in helmets. It also addresses challenges such as PCM encapsulation, and compatibility with other surrounding helmet materials. Furthermore, it also investigates the practical considerations for real-world applications of this PCM material.

Furthermore, the study describes innovative design solutions and methodologies for successfully incorporating PCMs into motorbike helmet liners. This will ensure optimum thermal function without degrading safety or comfort.

Overall, this case study provides a comprehensive investigation of the potential advantages, issues, and innovative methods associated with implementing paraffin wax PCMs into motorbike helmets.

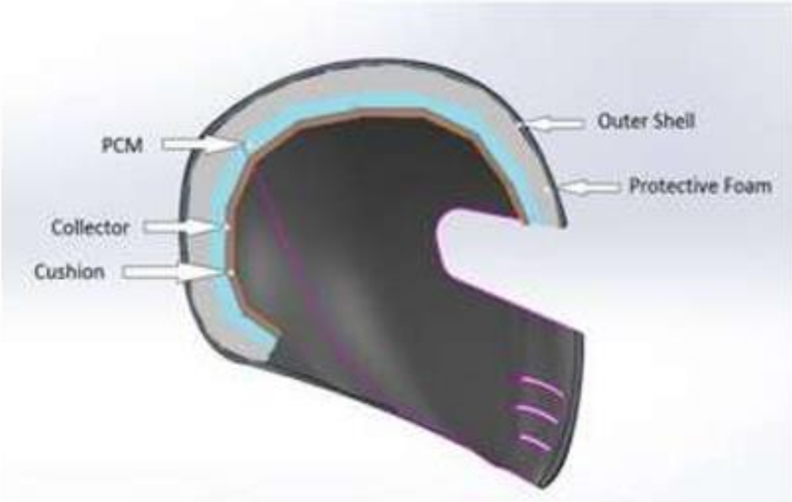


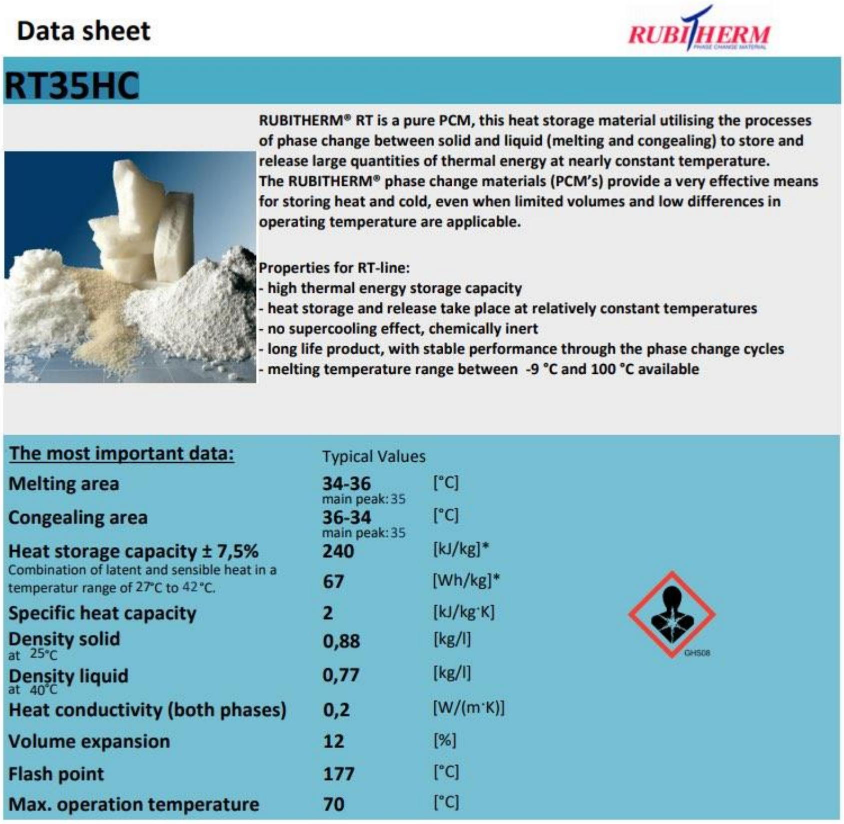
Figure 8. Setup of PCM in the helmet.

Figure 9. PCM in plastic pouches.

Figure 10. PCM with collector enclosed in fabric.

Simulation Setup Description

For our simulation, we chose a pure organic PCM, RT35HC, manufactured by RUBITHERM.



Data sheet

RUBITHERM

RT35HC

RUBITHERM® RT is a pure PCM, this heat storage material utilising the processes of phase change between solid and liquid (melting and congealing) to store and release large quantities of thermal energy at nearly constant temperature.
The RUBITHERM® phase change materials (PCM's) provide a very effective means for storing heat and cold, even when limited volumes and low differences in operating temperature are applicable.

Properties for RT-line:
- high thermal energy storage capacity
- heat storage and release take place at relatively constant temperatures
- no supercooling effect, chemically inert
- long life product, with stable performance through the phase change cycles
- melting temperature range between -9 °C and 100 °C available

The most important data:	Typical Values	
Melting area	34-36 main peak: 35	[°C]
Congealing area	36-34 main peak: 35	[°C]
Heat storage capacity ± 7,5%	240	[kJ/kg]*
Combination of latent and sensible heat in a temperatur range of 27°C to 42°C.	67	[Wh/kg]*
Specific heat capacity	2	[kJ/kg·K]
Density solid at 25°C	0,88	[kg/l]
Density liquid at 40°C	0,77	[kg/l]
Heat conductivity (both phases)	0,2	[W/(m·K)]
Volume expansion	12	[%]
Flash point	177	[°C]
Max. operation temperature	70	[°C]

Figure 11. Specifications of paraffin RT35HC.

This PCM can store and release large amounts of thermal energy and has a melting temperature of 34 to 36°C. These properties were found to be ideal for our helmet. After analysing the datasheet, we determined that this PCM would be the most suitable for our helmet and utilized Ansys to conduct simulations for practical testing.

We utilised the Fluid Flow (Fluent) analysis method for our simulations.

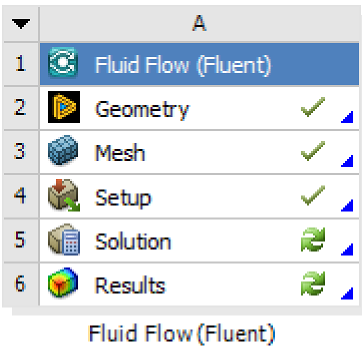


Figure 12. Fluid Flow (Fluent).

We designed our domain layers and created a mesh of size 5mm. Our mesh consisted of 16,224 Nodes and 12,500 Elements.

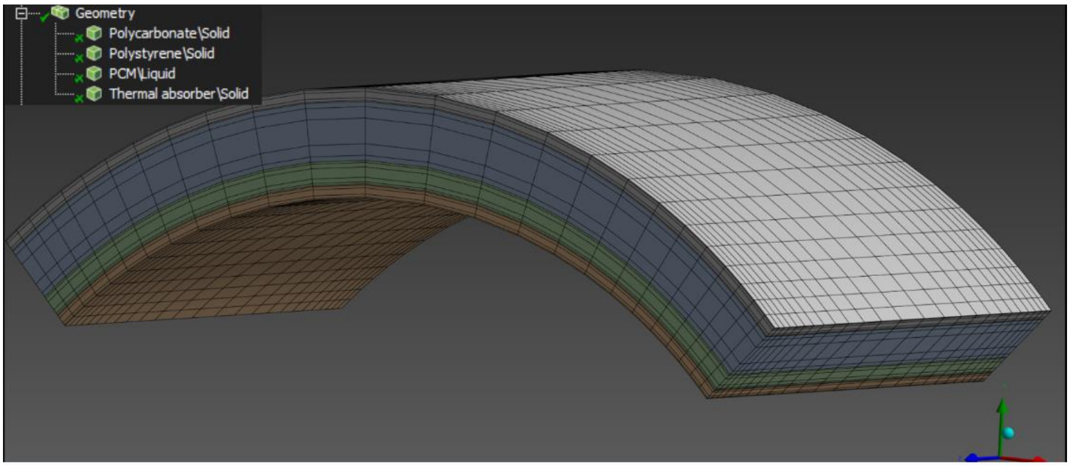


Figure 13. Domain layers and created a mesh of size 5mm.

Our simulation settings were as follows:

- Standard initialisation was chosen as it gives better control of the initial conditions.
- Least squares cell-based used for gradient for accuracy and computational efficiency.
- SIMPLE scheme as it is computationally quicker and overall computational cost can allow for performance of more iterations faster.
- PRESTO solves for the pressure on cell faces whereas standard and second Order interpolates them and gives a more physically accurate representation of the pressure field.
- Second order is used for energy, it is harder to converge the solution but more accurate.

Using the datasheet, we inputted the data into Ansys and set the properties of both the PCM and Polystyrene layers.

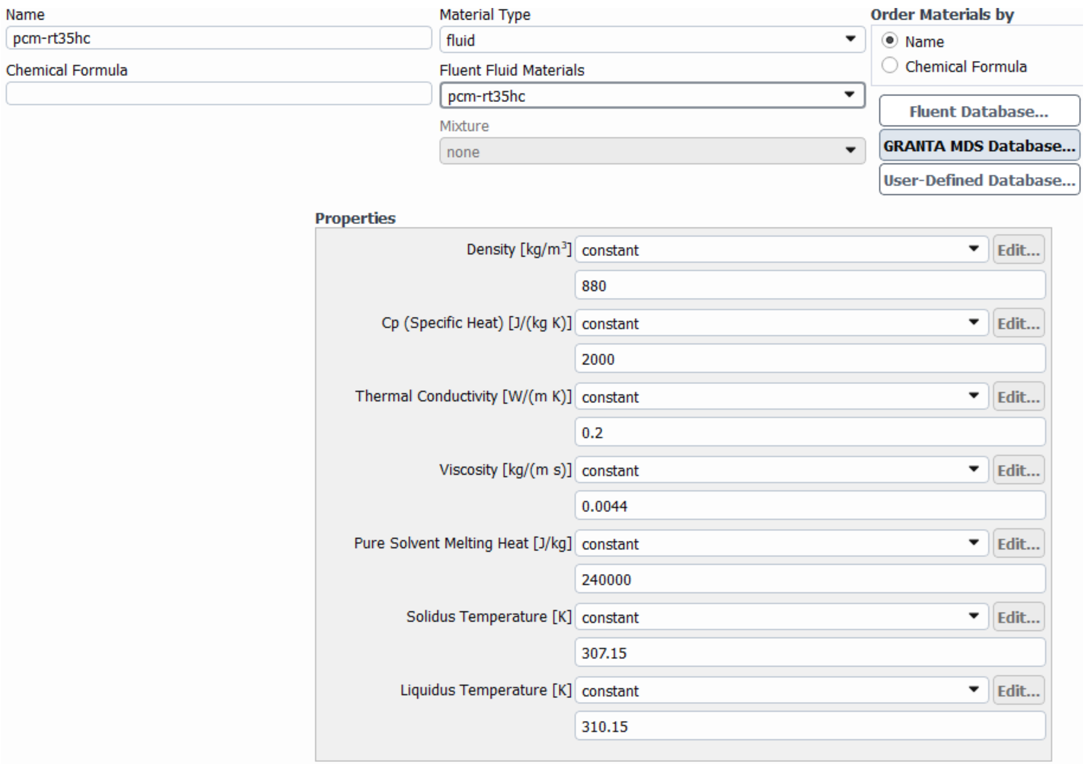


Figure 14. Ansys Data. Properties of both the PCM and Polystyrene layers.

Table 1. Specific heats, viscosities and thermal expansion coefficients of PCMs

Properties	Matter State	RT27 [11]	RT21 [11]	RT35HC [12]
C_p ($J.kg^{-1}.K^{-1}$)	Solid	2400	2100	2000
	Liquid	1800	4200	2000
μ ($kg.m^{-1}s^{-1}$)		3.4210^{-3}	2.810^{-3}	2.710^{-3}
β (K^{-1})		0.510^{-3}	0.510^{-3}	0.7610^{-3}

Figure 15. Specific heats and viscosities of PCMs sourced from (Kraiem., et al. 2019).

For the boundary conditions, we learned that according to various scientific articles mainly (Shennoy.S,. 2019 and Fitzgerald,. Et al. 2018) the heat flux between the human head and a motorcycle helmet in motion can vary between 115–260 W/m^2 at the parietal (back) zone of the head.

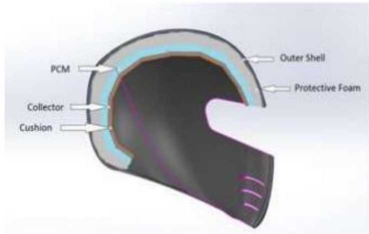


Figure 3- Setup of PCM in the helmet

1.2 Calculations

The cross-section of the helmet is as shown in figure 1. The PCM in the helmet gets heated up by 2 sources, i.e., the hotter air surrounding the helmet (ambient air), and the temperature of the head of the wearer. Generally, the heat generated by the head of the rider is 115 W/m², and hence about 15 W of heat is produced for an average size head.

Figure 16. Heat flux in a helmet. Source: (Shennoy.S,. 2019).

To find the heat flux at the boundary we used the equation for composite wall heat transfer as shown below.

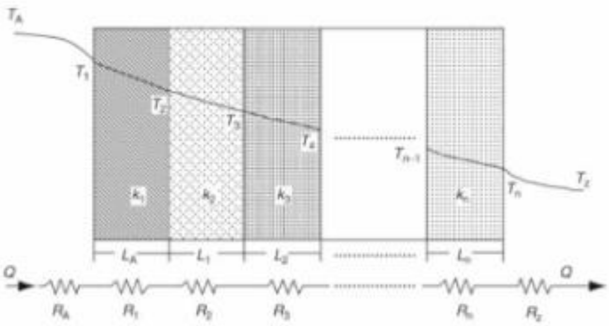


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Fig. 10. Thermal resistances in series in a composite wall.

$$Q = \frac{T_A - T_1}{\frac{1}{h_A A}} = \frac{T_1 - T_2}{\frac{L_1}{k_1 A}} = \frac{T_n - T_B}{\frac{1}{h_B A}} \qquad (41)$$

Figure 17. Equation for composite wall heat transfer. Source (Dincer., et al. 2019).

Our helmet design is composed of composite walls that comprise several layers of different materials with different properties and thicknesses. The rate of heat transfer is associated with the temperature difference and the thermal resistance of each layer. (Dincer., et al 2019)

$$Q = \frac{\Delta T}{\Sigma R}$$

$$\Sigma R = R1 + R1$$

$$R1(heat\ pad) = \frac{x}{k} = \frac{0.005}{0.29}$$

$$R2(PCM) = \frac{x}{k} = \frac{0.005}{0.2}$$

$$\therefore Q = \frac{310.15 - 303.15}{0.042} = 164.29W/m^2$$

Figure 18. Heat flux calculation.

Where x is the width of the boundary and k is the thermal conductivity of the material.

Applying the equation as shown resulted in a heat flux of 164 W/m^2 which corresponds to the research findings. This value was used as the boundary condition for the bottom layer.

We set the top layer or heat input temperature to 303K (29°C).

The simulations were run to find the volume fraction r liquid fraction in this case is the fraction of liquid per unit volume of the material, each cell within the computational domain is assigned to either a solidus, liquidus, or mushy state depending on the local cell temperature. During the experiment, liquid density varies due to liquid thermal expansion, while the solid density is constant. So, the liquid mass fraction is:

$$\gamma = \frac{m_{liq}}{m_{total}} = 1 - \frac{m_{\text{solid}}}{m_{\text{total}}} = 1 - \frac{\rho_{\text{PCM,Solid}} \forall_{\text{PCM,Solid}}}{\rho_{\text{PCM,Solid}} \forall_{\text{0,PCM,Solid}}} = 1 - \frac{\forall_{\text{PCM,Solid}}}{\forall_{\text{0,PCM,Solid}}}$$

Figure 19. Liquid mass fraction.

The simulations were also run to find the residual energy convergence, Residual convergence is achieved when the solution no longer varies significantly, signifying that the simulation has reached a stable state. This means that the solution has become accurate following significant iterations and is within the bounds of the convergence error criteria set in the monitors.

3D Printing – Chosen Design Explanation

In our sprint 4 presentation, we showcased our 3D-printed models of the Helmet and the Domain Layers. We utilised SolidWorks for designing our models.

Helmet

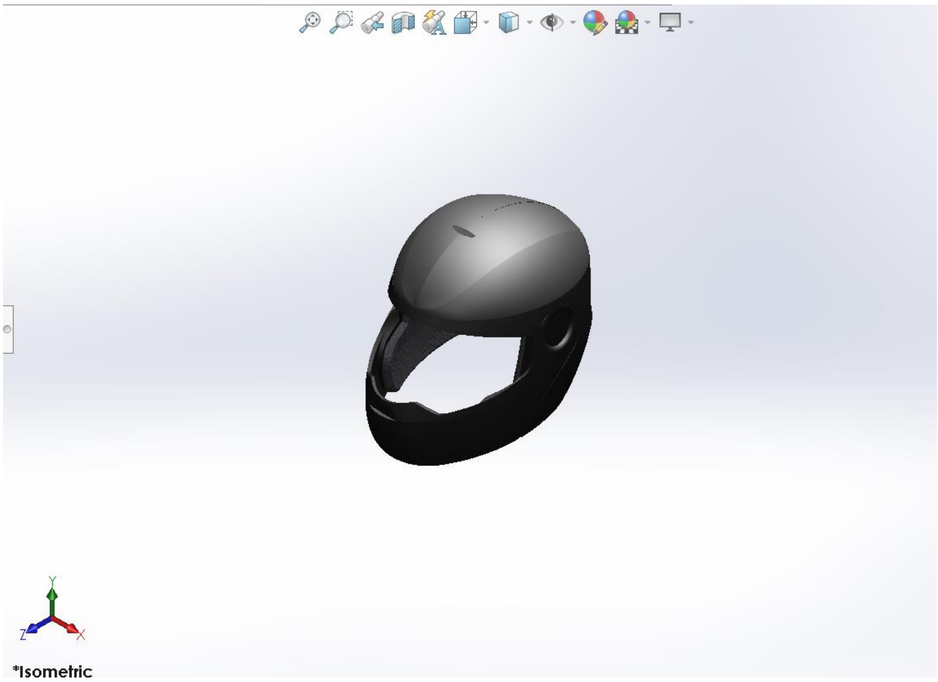


Figure 20. Isometric View of Helmet.

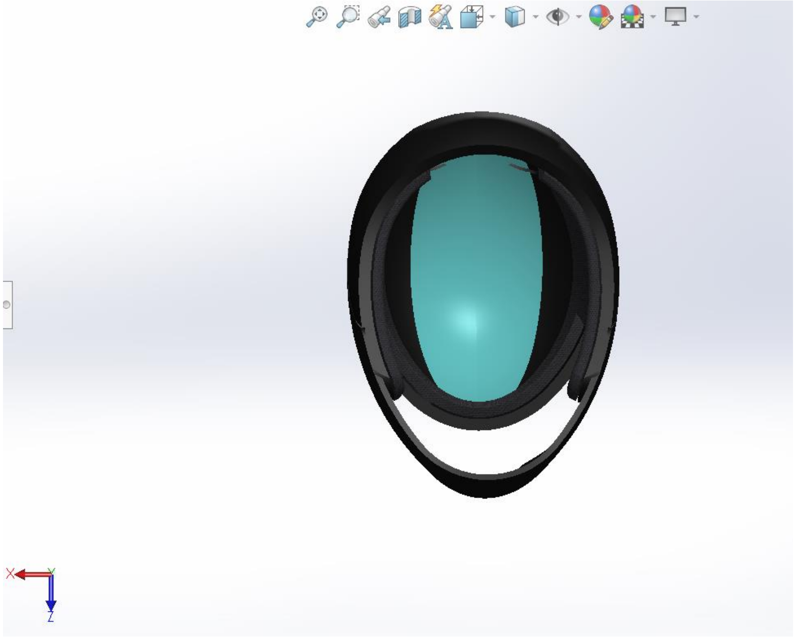


Figure 21. Location of PCM Domain.

Our helmet is designed to meet the standard dimensions of a motorcycle helmet. The dimensions of our helmet are 270x285x210mm (length x width x height). We incorporated our PCM Domain (Highlighted in Cyan) in the upper area of the helmet where there would be maximum contact with the user's head, which increased thermal conduction and effectiveness.

We 3D printed our helmet with a scale-down factor of 83%. This brought the dimensions down to 77.1x108.9x83.8mm (length x width x height).

In our first attempt at printing this model at ProtoSpace, we were presented with a low-quality print due to the constraints of ProtoSpace. We were not allowed to adjust any settings to increase the accuracy and finish of the print.

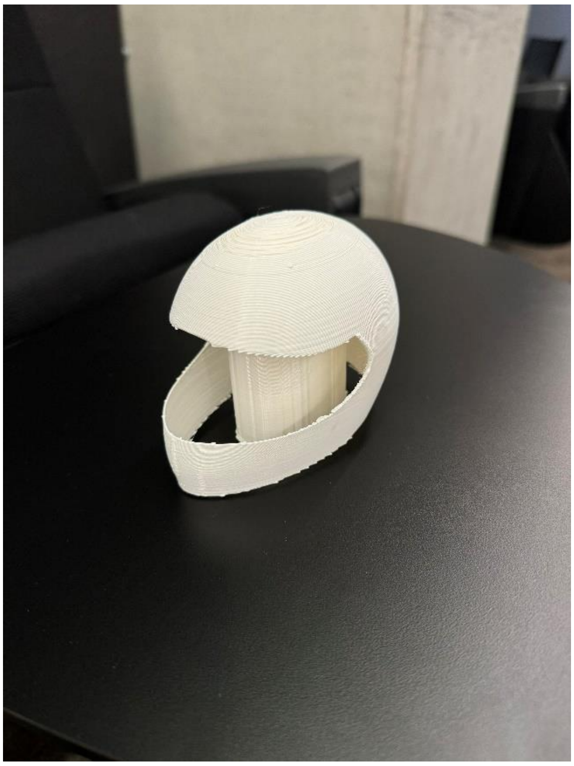


Figure 22. Helmet 3D Printed at ProtoSpace.

Due to the unsatisfactory results, we chose to 3D print our models privately. As a result, we were able to achieve greater print finishes and accuracy with custom settings.

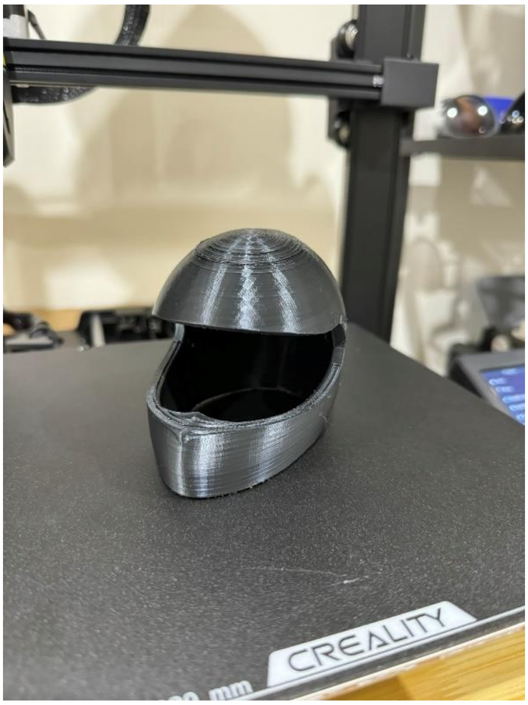


Figure 23. Helmet 3D Printed Privately.

This model took roughly 13 Hours to complete and used 20.8 meters of print filament. The total weight of our model was approximately 62 grams.

Figure 24. Side-by-side comparison.

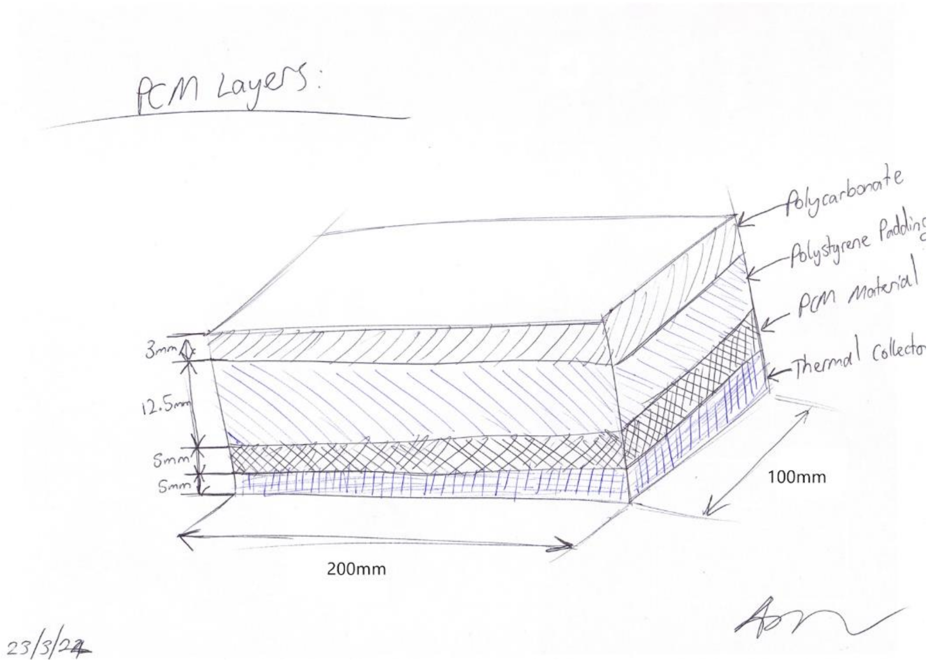


Figure 25. Hand Drawing of Domain Layers.

For our Domain Layer model, we incorporated a LEGO-style design where each layer would clip on top of each other, forming the complete domain. To ensure each layer was clipped in order, engraved labels were added to both the tops and sides of each layer to aid in assembly order.

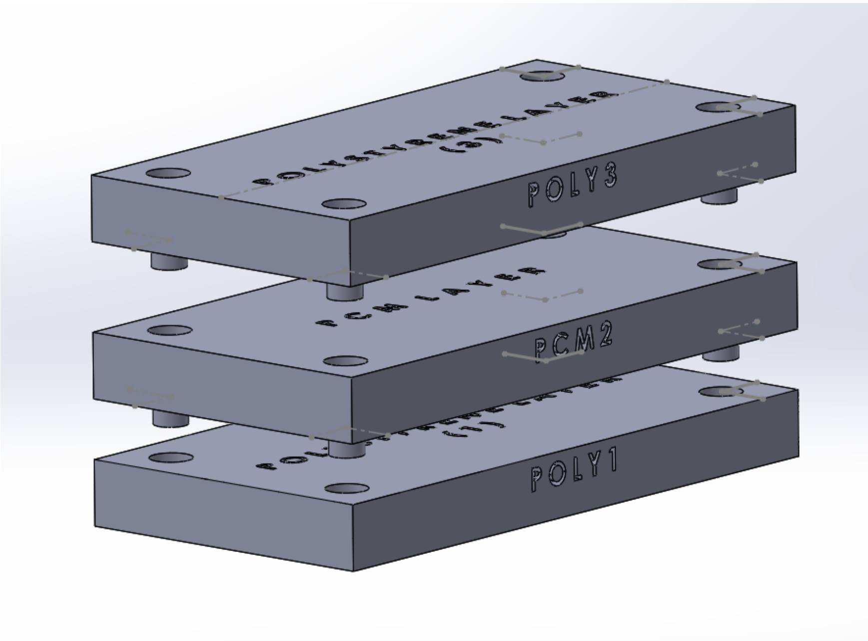


Figure 26. Exploded View of Domain.

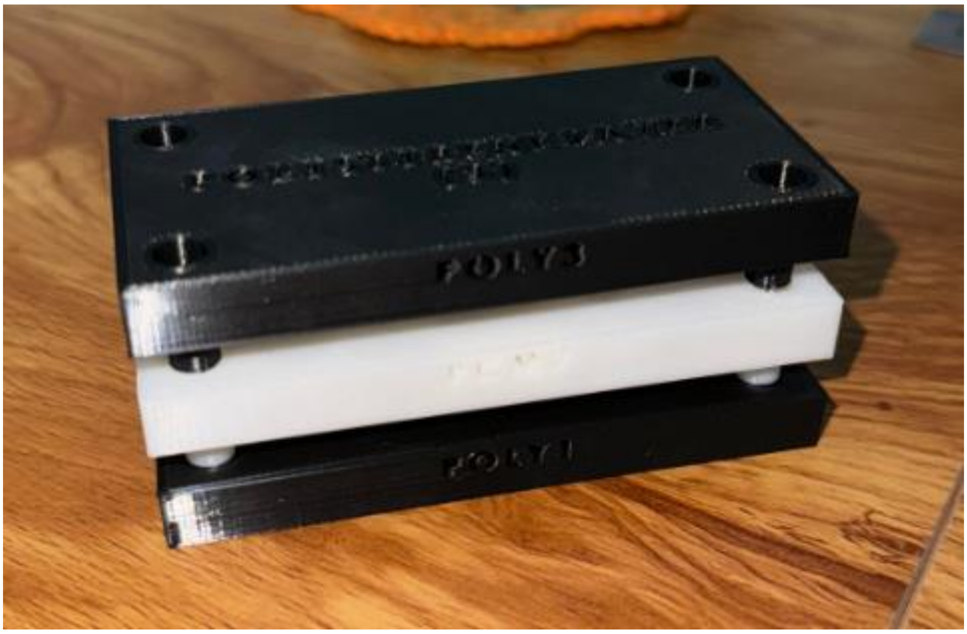


Figure 27. 3D Model (PCM layer in White).

Unfortunately, due to an error in dimensioning on SolidWorks, the holes were 0.5mm misaligned, resulting in the layers not clipping into each other, as designed. We were able to rectify this error however due to time constraints, we were unable to re-print the model in time for our Sprint 4 presentation. We also designed a more unique and realistic representation of our Domain.

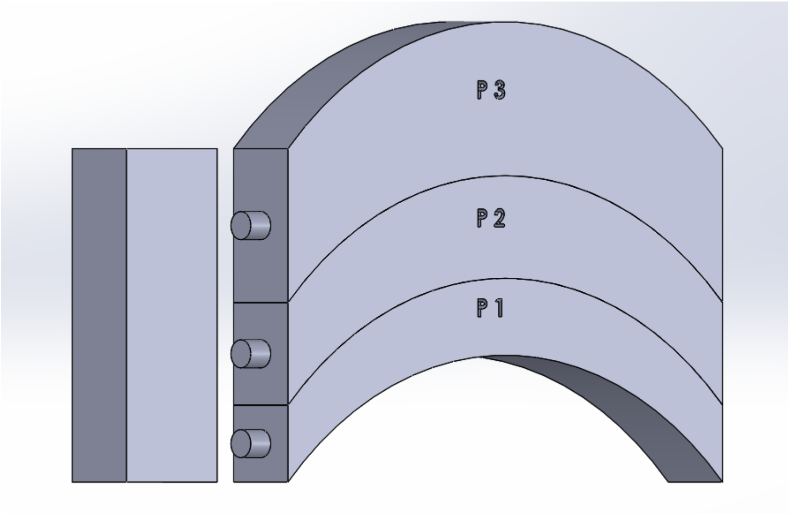


Figure 28. 3D CAD model of domain layers.

Figure 29. Fully assembled model.

This design accurately represented the helmet's curved shape. We incorporated a similar LEGO design where each layer simply stacks on top of each other and is securely combined with a connector block. Each layer had an engraving denoting its order and was uniquely shaped to prevent them from being stacked and connected incorrectly.

Results Discussion

Simulation Results and Analysis

The selection of the fine mesh for our simulations was substantiated by the residual graphs, which demonstrated satisfactory convergence, thereby ensuring the reliability of our computational results. The primary PCM used in our study, RT 37-HF, exhibited a complete phase change after 2955 seconds (approximately 49 minutes). This was evidenced by the volume fraction graph, which indicated the transition of the PCM from a solid to a liquid state.

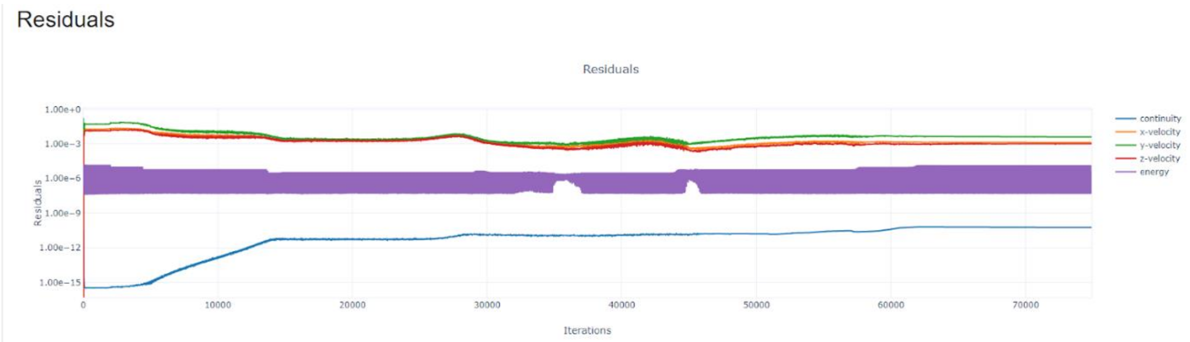


Figure 30. The residual graph of RT 37-HF.

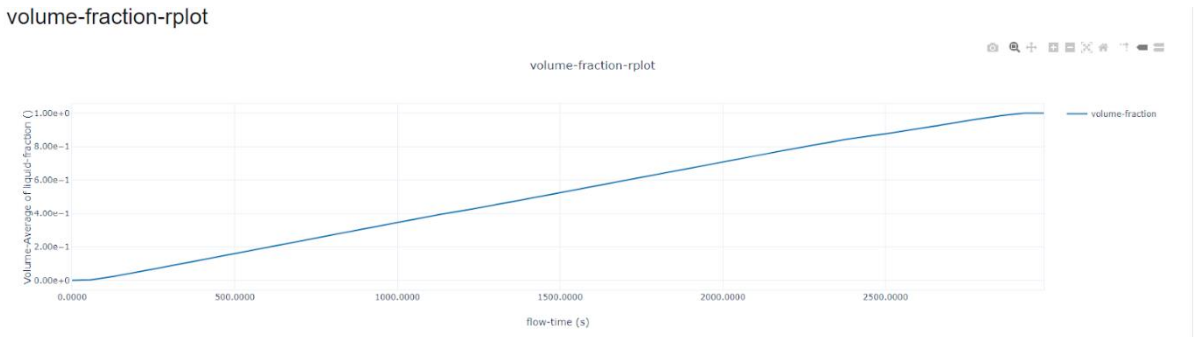


Figure 31. The volume fraction graph of RT 37-HF.

The liquid fraction contours of RT 37-HF at 2000 seconds depicted the initiation of melting from the central region, with temperature gradients extending towards the periphery. By 2955 seconds, the PCM had entirely melted, achieving a liquid fraction value of one. The corresponding temperature contour analysis revealed a notable increase in temperature, particularly after the melting point was reached, confirming the effective absorption and distribution of thermal energy by the PCM.

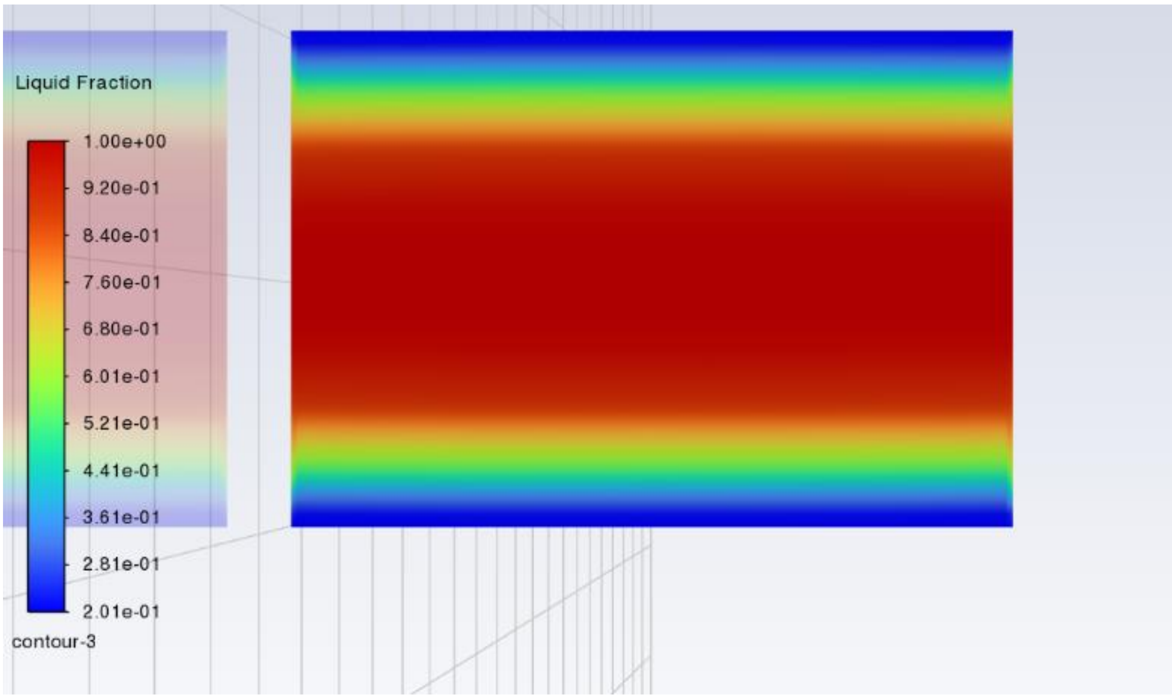


Figure 32. The liquid fraction contours of RT 37-HF at 2000 seconds.

Figure 33. The liquid fraction contours of RT 37-HF at 2955 seconds.

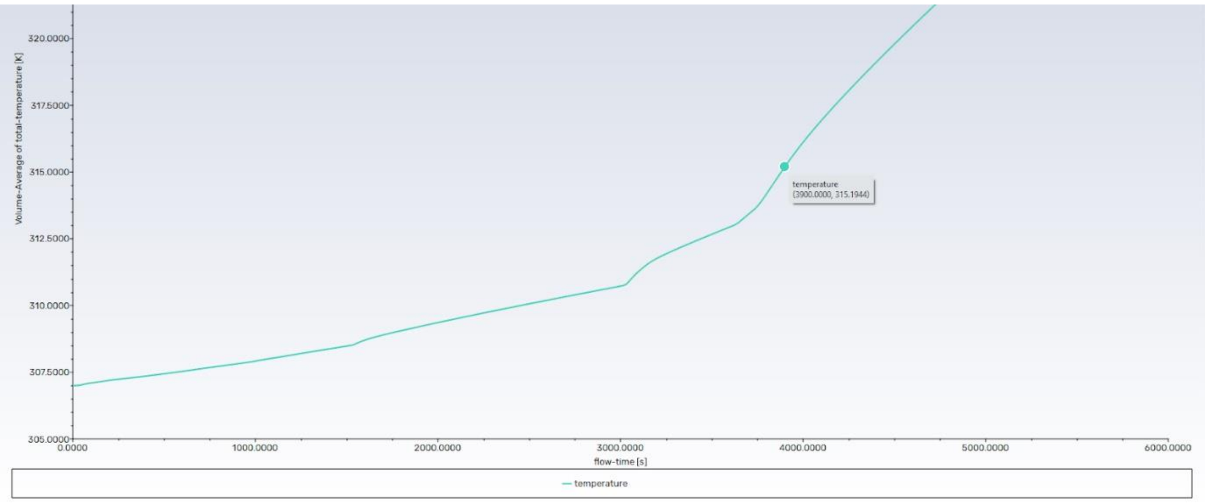


Figure 34. The total temperature graph of RT 37-HF.

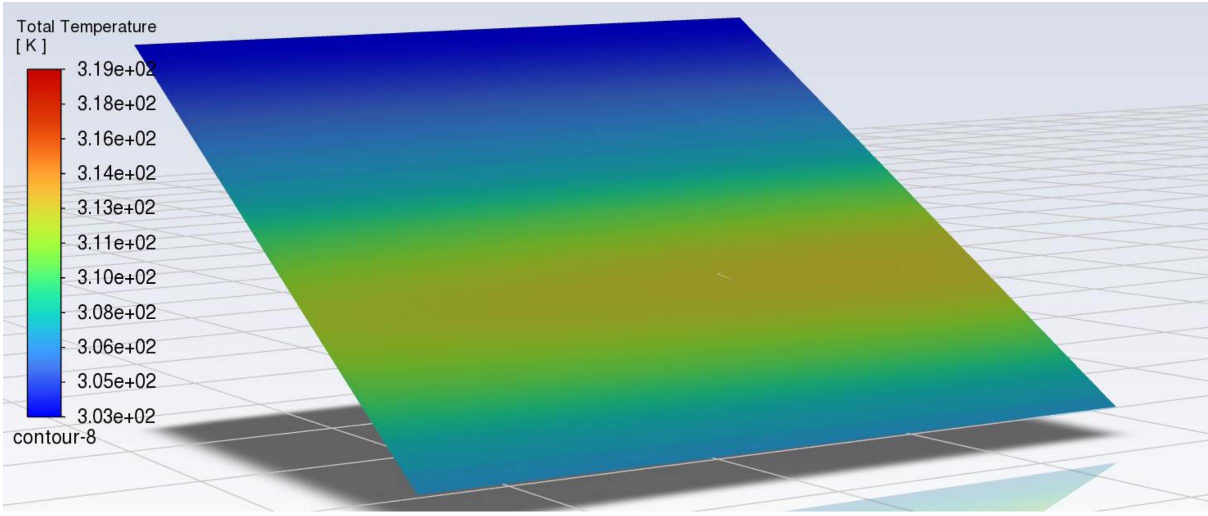


Figure 35. The temperature contours of RT 37-HF at 2955 seconds.

Upon transitioning to a new PCM, PlusICE S32, the simulations maintained all conditions constant except for the PCM material itself. The results indicated that PlusICE S32 required 4680 seconds (approximately 78 minutes) to achieve complete melting. This extended melting duration, as compared to RT 37-HF, suggests a different thermal management capability, potentially offering longer cooling periods.

PCM — PlusICE Hydrated Salt (S) Range

PCM Type	Phase Change Temperature		Density		Latent Heat Capacity		Volumetric Heat Capacity		Specific Heat Capacity		Thermal Conductivity		Maximum Operating Temperature	
	(°C)	(°F)	(kg/m³)	(lb/ft³)	(kJ/kg)	(Btu/lb)	(MJ/m³)	(Btu/ft³)	(kJ/kgK)	(Btu/lb°F)	(W/mK)	(Btu/Hr ft² °F)	(°C)	(°F)
S8	8	46	1,475	92	130	56	192	5,147	1.9	0.45	0.44	0.25	60	140
S10	10	50	1,470	92	170	73	250	6,707	1.9	0.45	0.43	0.25	60	140
S13	13	55	1,515	95	150	65	227	6,099	1.9	0.45	0.43	0.25	60	140
S15	15	59	1,510	94	180	77	272	7,295	1.9	0.45	0.43	0.25	60	140
S17	17	63	1,525	95	155	67	236	6,344	1.9	0.45	0.43	0.25	60	140
S18	18	64	1,520	95	145	62	220	5,916	1.9	0.45	0.43	0.25	60	140
S19	19	66	1,520	95	175	75	266	7,139	1.9	0.45	0.43	0.25	60	140
S20	20	68	1,530	96	195	84	298	8,008	2.2	0.52	0.54	0.31	60	140
S21	21	70	1,530	96	220	95	337	9,034	2.2	0.52	0.54	0.31	60	140
S22	22	72	1,530	96	215	93	329	8,829	2.2	0.52	0.54	0.31	60	140
S23	23	73	1,530	96	200	86	306	8,213	2.2	0.52	0.54	0.31	60	140
S24	24	75	1,530	96	180	77	275	7,392	2.2	0.52	0.54	0.31	60	140
S25	25	77	1,530	96	175	75	268	7,186	2.2	0.52	0.54	0.31	60	140
S27	27	81	1,530	96	185	80	283	7,597	2.2	0.52	0.54	0.31	60	140
S32	32	90	1,460	91	220	95	321	8,621	1.9	0.45	0.51	0.29	60	140
S34	34	93	2,100	131	140	60	294	7,891	2.1	0.5	0.52	0.3	70	158

Figure 36. Specifications of PlusICE S32.

volume-fraction-rplot

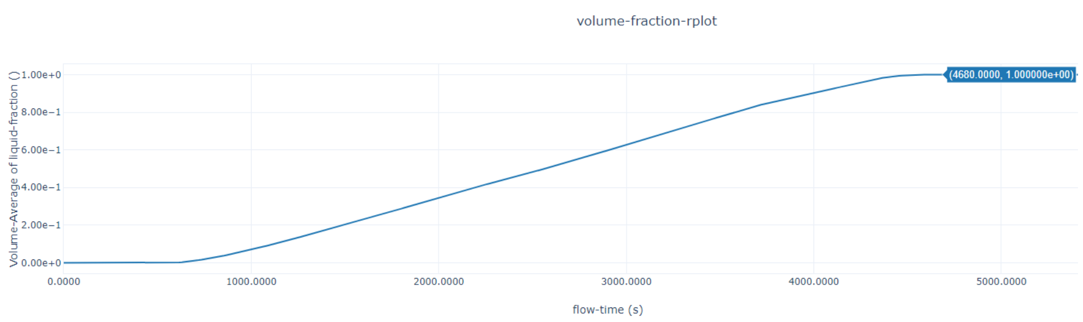


Figure 37. The volume fraction graph of PlusICE S32.

The volume fraction graph for PlusICE S32 supported these findings, displaying a gradual phase transition over the specified duration. The liquid fraction contours at 2000 seconds showed minimal melting, with temperature increments originating from the core. At 4680 seconds, the PCM had fully transitioned to the liquid state, as illustrated by the corresponding liquid fraction and temperature contours.

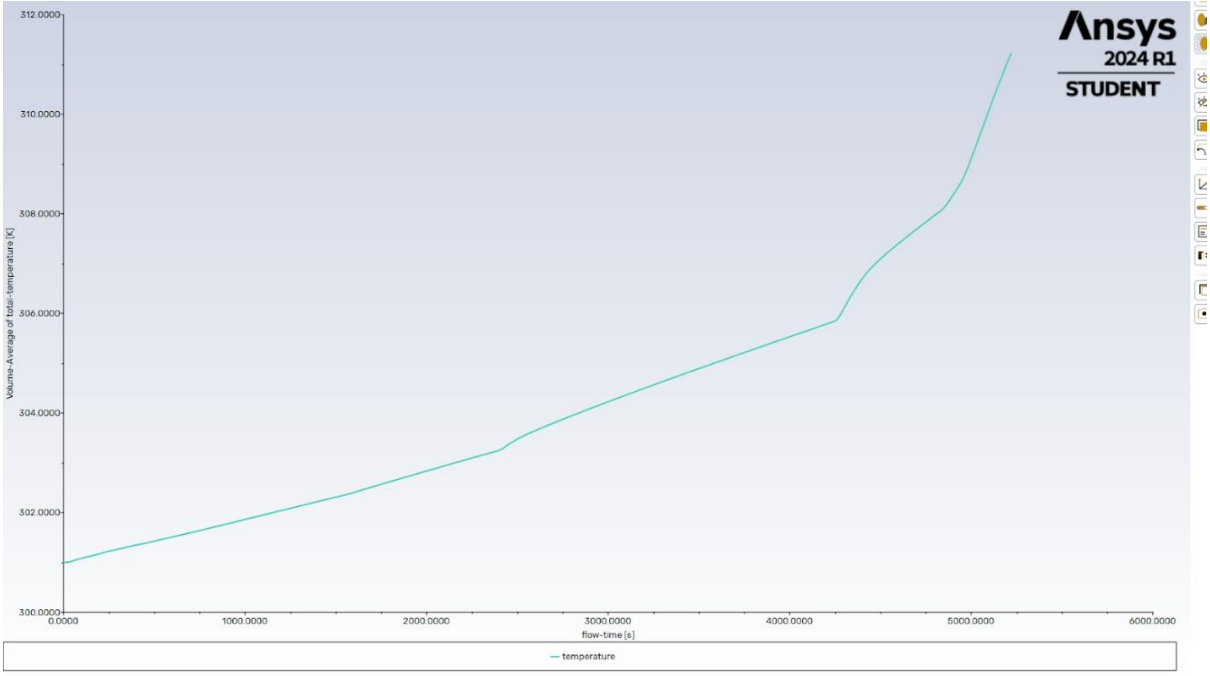


Figure 38. The temperature graph of PlusICE S32.

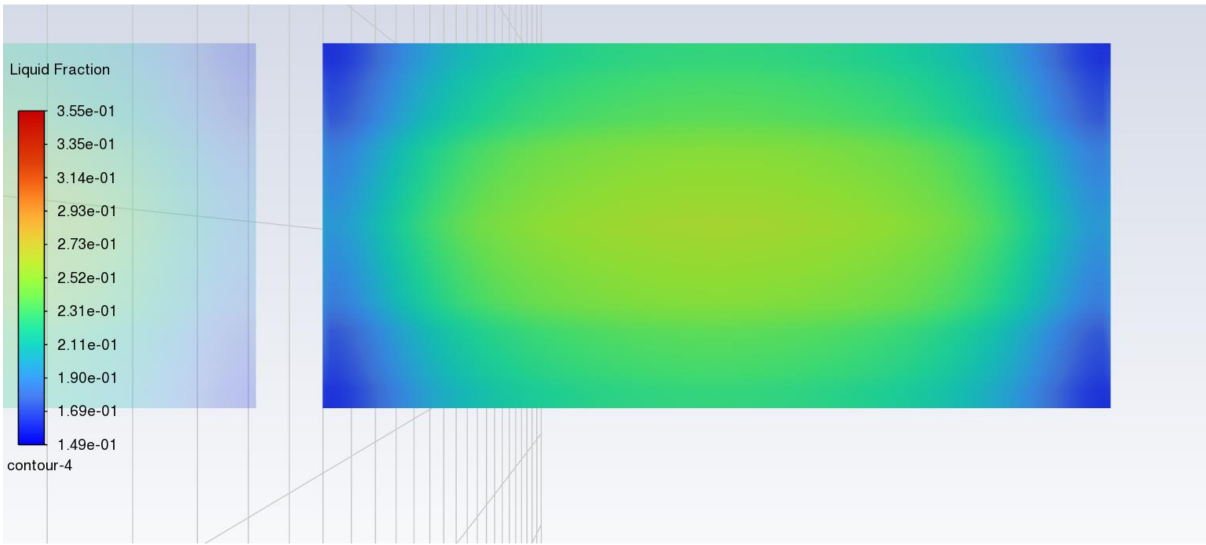


Figure 39. The liquid fraction contours of PlusICE S32 at 2000 seconds.

Figure 40. The liquid fraction contours of PlusICE S32 at 4680 seconds.

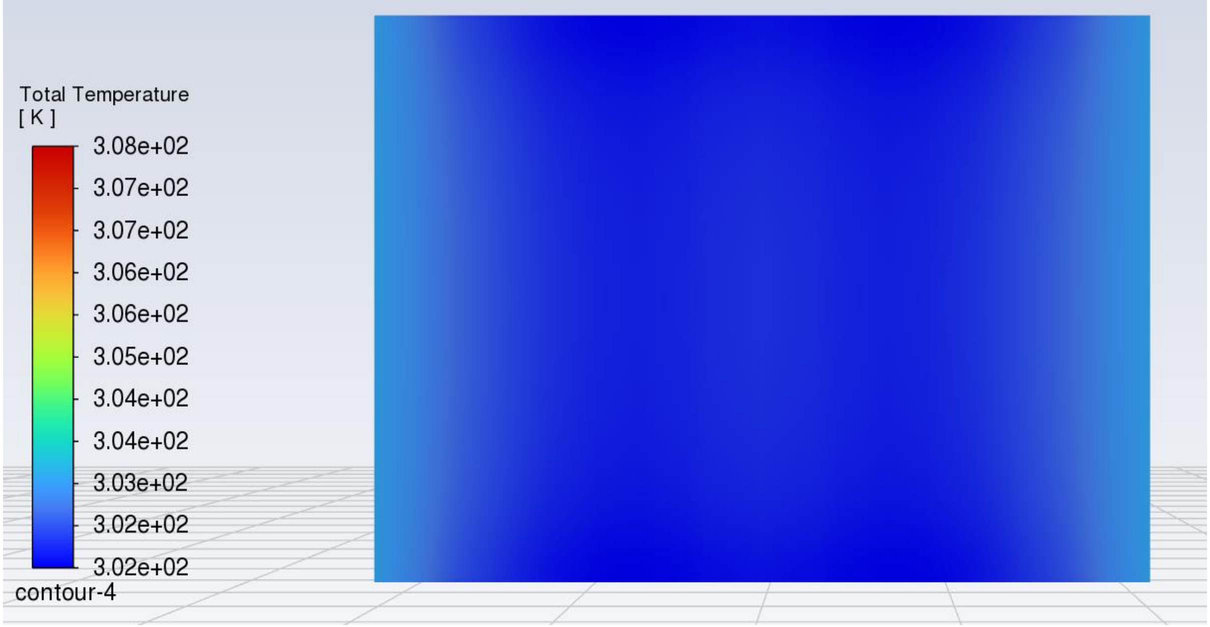


Figure 41. The total temperature contours of PlusICE S32 at the beginning.

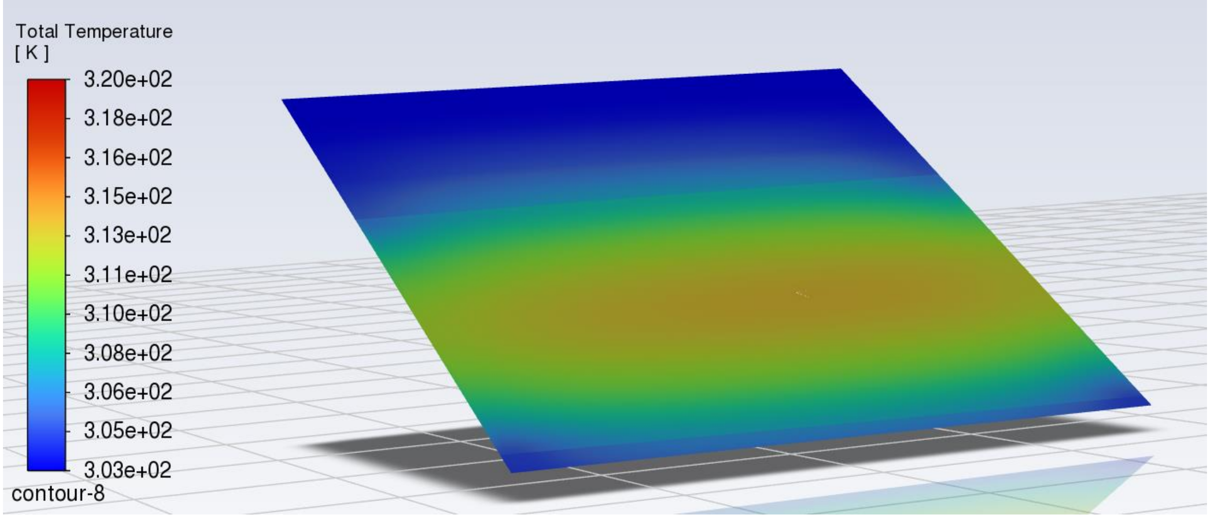


Figure 42. The total temperature contours of PlusICE S32 at 4680 seconds.

The total temperature graphs for both PCMs demonstrated a sharp increase in temperature post-melting, emphasising the critical role of the PCM's thermal absorption capacity in regulating helmet temperatures.

Implications for PCM Helmet Design

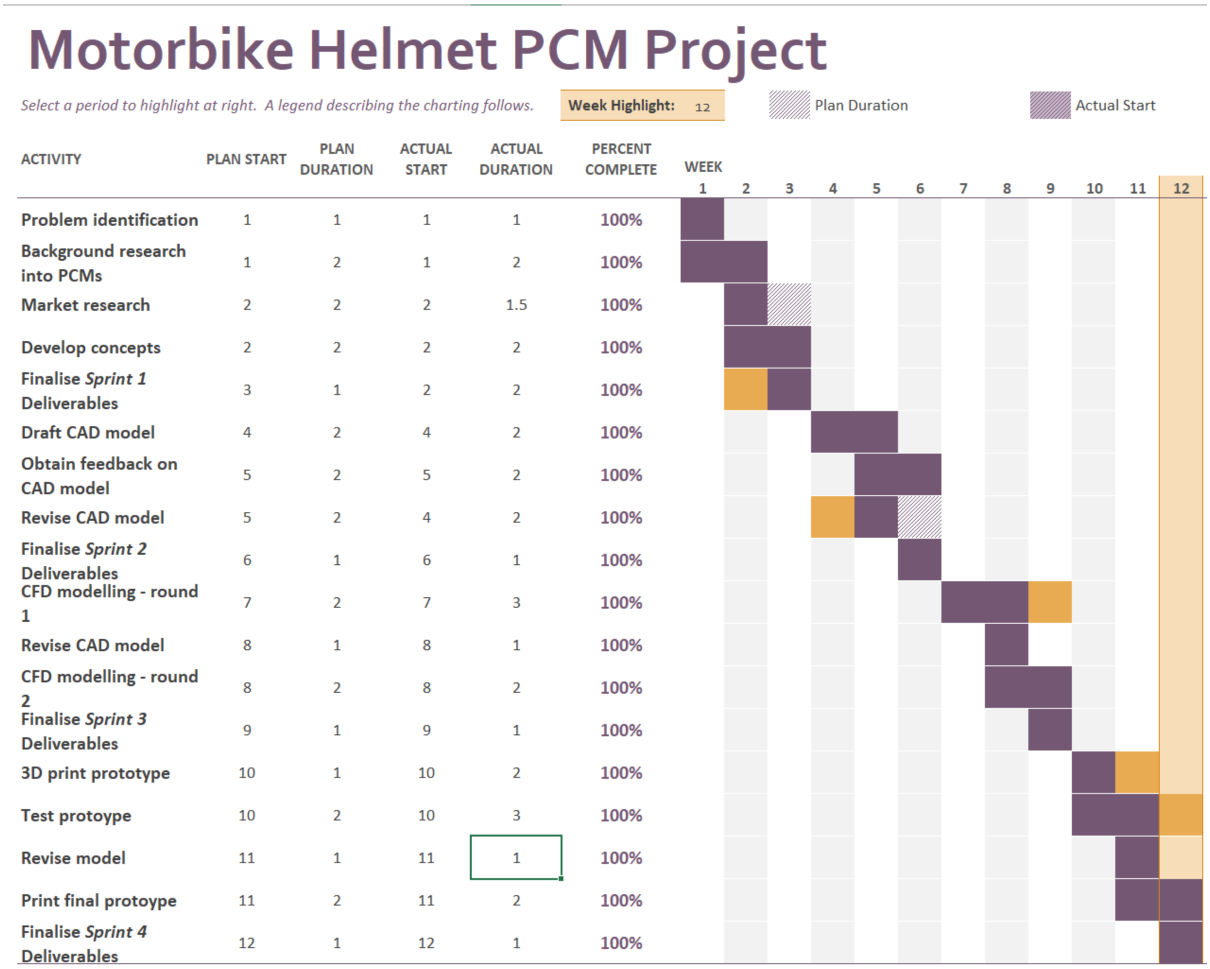
The comparative analysis of RT 37-HF and PlusICE S32 highlights the importance of PCM selection based on the desired cooling duration and thermal management requirements. RT 37-HF, with its quicker melting time, is suitable for shorter, high-intensity cooling needs, whereas PlusICE S32 provides extended cooling, making it ideal for prolonged use in moderate conditions.

These findings underscore the necessity for a tailored approach in PCM integration, ensuring that the selected material aligns with the specific thermal demands of the application. The successful simulation and post-processing results validate the feasibility of using PCMs for enhanced thermal management in motorcycle helmets, paving the way for further optimisation and practical implementation.

Project Management

Project Gannt Chart

A project Gantt chart was developed and utilised as the primary tool to manage tasks and deadlines for this project. This tool proved invaluable when the workload increased during the later Sprints.



ACTIVITY	PLAN START	PLAN DURATION	ACTUAL START	ACTUAL DURATION	PERCENT COMPLETE
Problem identification	1	1	1	1	100%
Background research into PCMs	1	2	1	2	100%
Market research	2	2	2	1.5	100%
Develop concepts	2	2	2	2	100%
Finalise *Sprint 1* Deliverables	3	1	2	2	100%
Draft CAD model	4	2	4	2	100%
Obtain feedback on CAD model	5	2	5	2	100%
Revise CAD model	5	2	4	2	100%
Finalise *Sprint 2* Deliverables	6	1	6	1	100%
CFD modelling - round 1	7	2	7	3	100%
Revise CAD model	8	1	8	1	100%
CFD modelling - round 2	8	2	8	2	100%
Finalise *Sprint 3* Deliverables	9	1	9	1	100%
3D print prototype	10	1	10	2	100%
Test protoype	10	2	10	3	100%
Revise model	11	1	11	1	100%
Print final protoype	11	2	11	2	100%
Finalise *Sprint 4* Deliverables	12	1	12	1	100%

Figure 43. Project Gantt chart.

Group Hierarchy

Feedback from Sprint 1 was implemented, and a group hierarchy was established. This allowed us to delegate tasks efficiently and effectively as our roles were clearly defined.

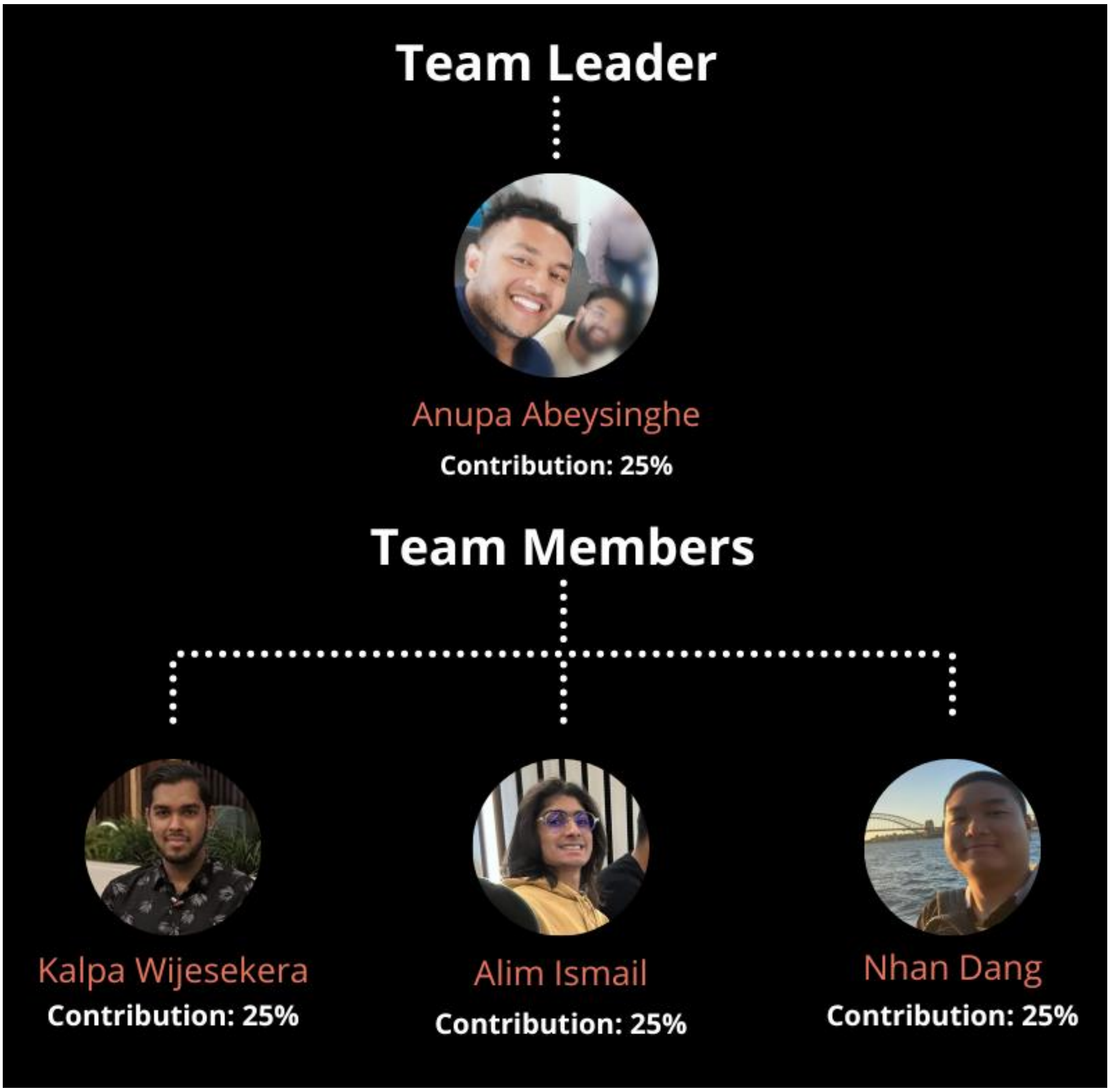


Figure 44. Group 24 hierarchy.

References

Fitzgerald, S., Atkins, H., Leknys, R., Kelso, R. (2018) *A Thermal Test System for Helmet Cooling Studies. Proceedings*. 2. 272. 10.3390/proceedings2060272.

Cornelis P. Bogerd, Jean-Marie Aerts, Simon Annaheim, Peter Bröde, Guido de Bruyne, Andreas D. Flouris, Kalev Kuklane, Tiago Sotto Mayor, René M. Rossi, (2015) *A review on ergonomics of headgear: Thermal effects,* International Journal of Industrial Ergonomics, Volume 45, 2015, Pages 1-12, ISSN 0169-8141, https://doi.org/10.1016/j.ergon.2014.10.004

Adil A.M. Omara, Abubaker A.M. Mohammedali, R. Dhivagar, (2023) *Helmets cooling with phase change materials: A systematic review*, Journal of Energy Storage, Volume 72, Part C, 108555, ISSN 2352-152X, https://doi.org/10.1016/j.est.2023.108555.

Arumugam, Chelliah & Karthick, B & Vimalkodeeswaran, A & Hariram, V. (2015). H*elmet cooling system using phase change material for long drive.* Journal of Engineering and Applied Sciences. 10.

F.L. Tan, S.C. Fok, 2006 *Cooling of helmet with phase change material,* Applied Thermal Engineering, Volume 26, Issues 17–18, Pages 2067-2072, ISSN 1359-4311, https://doi.org/10.1016/j.applthermaleng.2006.04.022

Sharifpour, E & Farrokhnia, Mohammadreza & Golzar, Farzin & Karimi, Gholamreza. (2012). *Improving thermal comfort of motorcycle helmet by using phase change materials.*

Shenoy, S. (2018). Integration of Phase Change Material in Helmets. International Research Journal of Engineering and Technology (IRJET), Volume 6, Issue 5.